

Tok Essay

Prompt 3.

Is the power of knowledge determined by the way in which the knowledge is conveyed? Discuss with reference to mathematics **and one other** area of knowledge.

Word count: 0000

Introduction

Let us define the power of knowledge as follows:

1. Knowledge that is persuasive is powerful.
2. Knowledge that influences decisions of policymakers is powerful.
3. Knowledge that affects the understanding of other knowledge is powerful.

1 Fermat's Last Theorem

The way in which knowledge is conveyed: From a figure of authority. A rigorous proof is the backbone of mathematics. Authority is useless (or has very little use) in the power of mathematical knowledge. Fermat, the greatest mathematician at his time, claims he has proven his last theorem $a^n + b^n = c^n$. Yet, before Wiles's proof and verification, the theorem is not treated as much more plausible than, say, nonexistence of odd perfect numbers.

2 Euclid's definition of a point

To anyone who has lived in a modern American city (except Boston) at least one of the underlying ideas of Descartes' analytic geometry will seem ridiculously evident. Yet, it took mathematicians all of two thousand years to arrive at this simple thing.

Bell, pp. 123

Really, it seems obvious to us that to describe points on a two-dimensional plane we may use *two dimensions*, the horizontal and the vertical. This makes it rather surprising

that such a representation was not developed any sooner than the fifteenth century, such as by the Ancient Greeks who are especially known for their study of geometry.

For argument's sake, let us assume that the Greeks noticed this idea, but for some reason which we will explore, it was never popularized or documented at the time. After all, the basic idea was no more than just comparing the widths and heights of rectangles who share a corner:

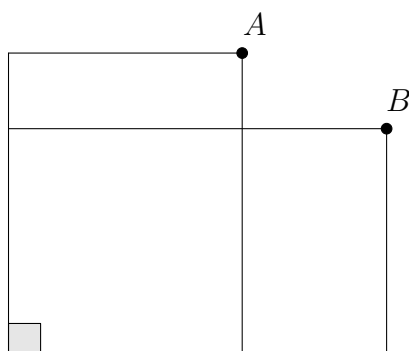


Figure 1: What the Ancient Greeks might have drawn.

It is known that the Ancient Greeks were quite capable in the study of the lengths in geometrical configurations, with examples like Archimedes' approximation of π . In fact, for the task of computing the distance between points A and B in **Figure 1**, we would use none other than the *Pythagorean* theorem, a testament to the Greeks' geometrical development.

Yet they never achieved the level of abstraction attained by the cartesian plane, which would turn out to be a very powerful tool. It could reduce the proof of or disproof of any claim to a purely mechanical routine that required no spatial intuition, much unlike the classical methods that relied on studying the diagram.

Notwithstanding the difference in available mathematical knowledge, I argue that a reason for the delay in the invention of analytic geometry is the way in which mathematical knowledge was conveyed in the ancient times and at Descartes' time. Specifically,

Ancient Greeks studied mathematics through spoken persuasion and with reference to geometrical diagrams and construction, differing from the modern approach of symbolic abstraction, which was already popularized at Descartes' time.

Because of the spoken tradition of presenting mathematical knowledge, there is an inherent risk of overlooking certain important details. Consider Euclid's definition of a point as an example. In his list of definitions in Book 1, Euclid writes:

Σημεῖόν ἐστιν, οὗ μέρος οὐθέν.

A point is that of which there is no part.

Such a definition might be acceptable at the time. Euclid's students likely did not challenge this definition, because it states what is intuitively the distinction between a point and all other geometrical objects. You can always divide a curve or a solid into parts, but there is no way to do this to a point. Moreover, this definition was, at the time, rather inescapable: Another characteristic of a point is that it has only position. Yet this risks a circular argument, since positions are simply points.

The oral tradition is the ultimate cause for this weakness of the Ancient Greek, since obviously for the sake of persuasion such an intuitively obvious definition is preferred, over some convoluted way of describing a point as side lengths of rectangles as drawn in **Figure 1**. This is a significant reason why Euclid chose not to define a point as 'represented by the pair of numbers (x, y) '.

For this reason, Ancient Greek geometers would never uncover the full power of their geometrical knowledge, which lies in Descartes' invention of the *coordinate system* that would unite geometry with algebra. The Pythagorean theorem was a way to compute hypotenuses of right triangles, but it would not become *The Distance Formula* until Descartes.

3 Hans Rosling's Gapminder

To scholars, decisions may seem ignorant; Yet there is a gap between scholars and Leaders, likely because how knowledge is conveyed. It's not enough that knowledge exists somewhere; conveying it is equally critical for the power of this knowledge to manifest.

I argue that certain knowledge (especially outside of mathematics), can only demonstrate its power when conveyed in a specific way. (or, if this is too strong, then that the way in which knowledge is conveyed amplifies its power significantly.)

Gapminder is considered successful, not for introducing new knowledge, but instead for propagating existing knowledge to a wider audience. Surely intuition correction is powerful.

It is rhetorical power. The site even tells you 'your answer is incorrect, just like 70% others.'

4 Whatever in the World, what even is the AOK???

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5 conclusion?

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